

NETWORK TRAFFIC ANALYSIS

PCAP INVESTIGATION ANALYSIS

IP ADDRESS NETWORK HEADERS

CYBERSECURITY RESEARCH

TASK 1 — NETWORK TRAFFIC ANALYSIS

1. Host Identification

The primary internal host observed in the supplied capture is 192.168.0.7. It is the main source of the investigated DNS, HTTP, TCP and encrypted web traffic.

2. Protocols Observed

- DNS
- TCP
- HTTP
- TLS/HTTPS
- ARP
- ICMP
- QUIC/UDP

The capture therefore contains both ordinary local-network activity and Internet-facing communications. Protocol interpretation below is based on the supplied Wireshark evidence.

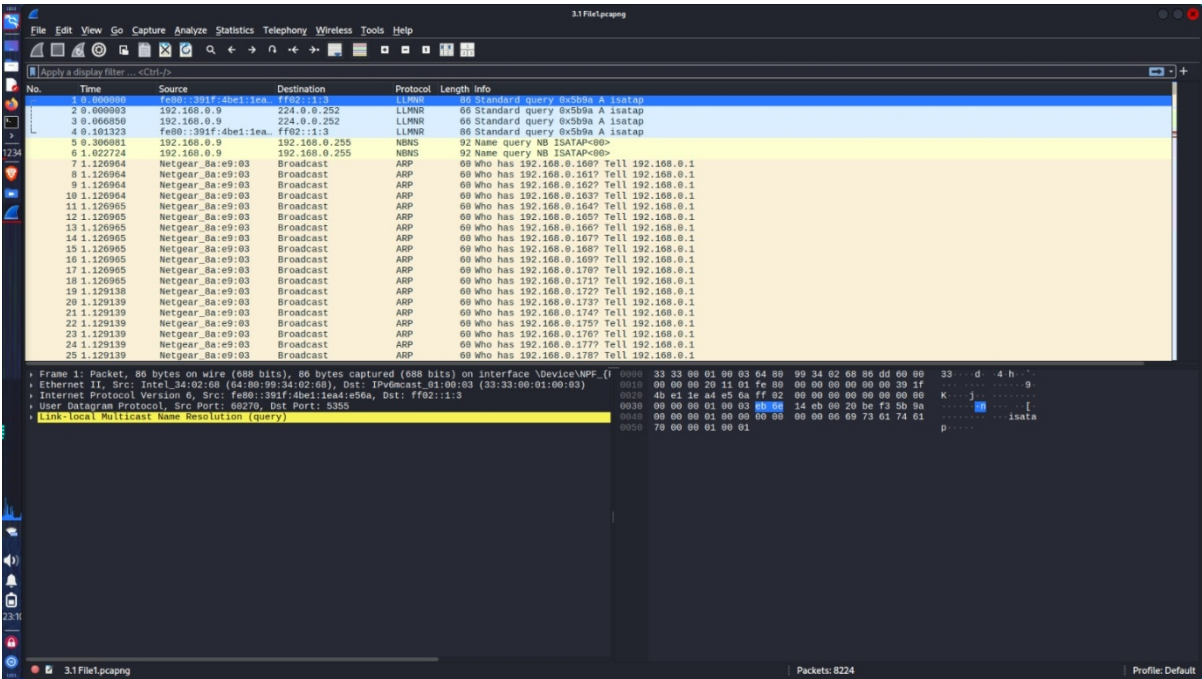


Figure 1. Main Wireshark packet view showing the traffic capture, no filter applied.

3. Important IP Addresses

The following addresses are directly visible in the supplied packet evidence:

Address	Role / Observation	Evidence context
192.168.0.7	Primary internal host	Source of investigated traffic
68.105.28.11	DNS server	DNS queries/responses
216.58.195.35	External destination	TCP/HTTPS connection
173.194.219.105	External destination	QUIC traffic
199.192.165.120	External destination	HTTP traffic
72.246.125.34	External destination	HTTP/web traffic

204.160.114.75	External destination	TLS/HTTPS traffic
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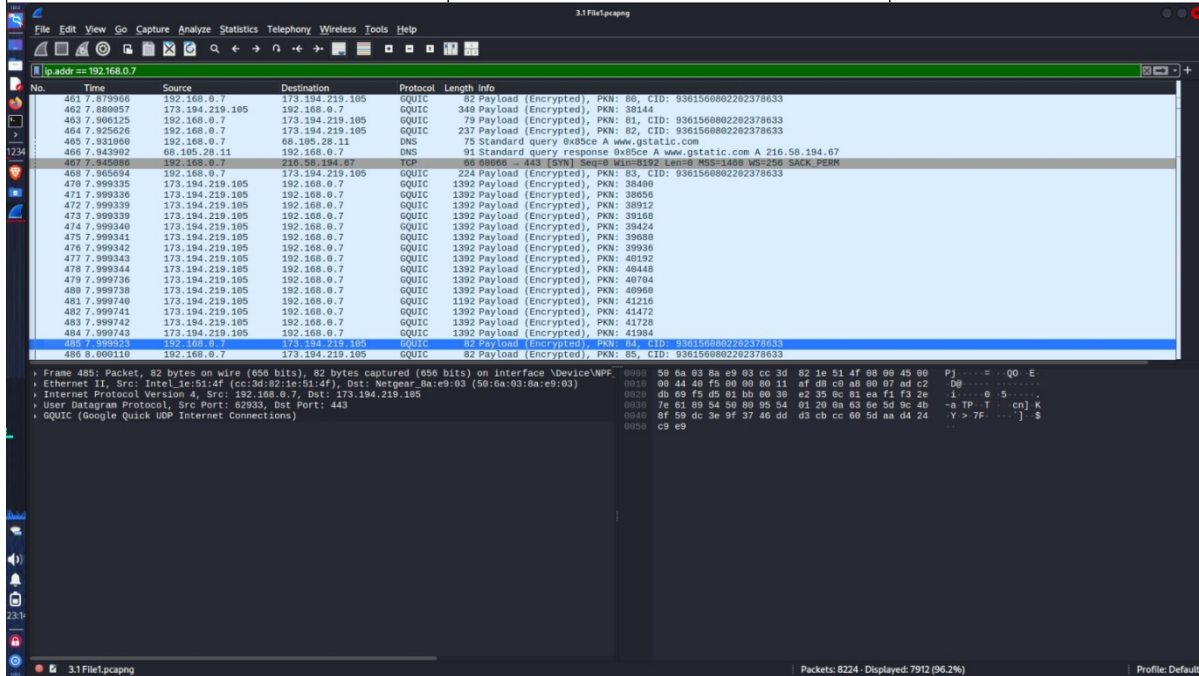


Figure 2. Wireshark filter `ip.addr == 192.168.0.7`, isolating traffic associated with the internal host.

4. Important Ports

- 53 — DNS
- 80 — HTTP
- 443 — HTTPS/TLS and other encrypted web traffic shown in the capture
- 60074 and other ephemeral client ports are dynamically assigned source ports used for individual TCP sessions

The capture also shows UDP-based QUIC traffic to port 443, demonstrating that modern web applications may use encrypted transport over UDP rather than TCP.

5. DNS Findings

DNS traffic translates domain names into IP addresses before or alongside application connections. The supplied capture shows DNS queries from 192.168.0.7 to the DNS server 68.105.28.11, including requests for `www.google.com` and `www.udemy.com`. DNS responses provide the destination address information used by the host.

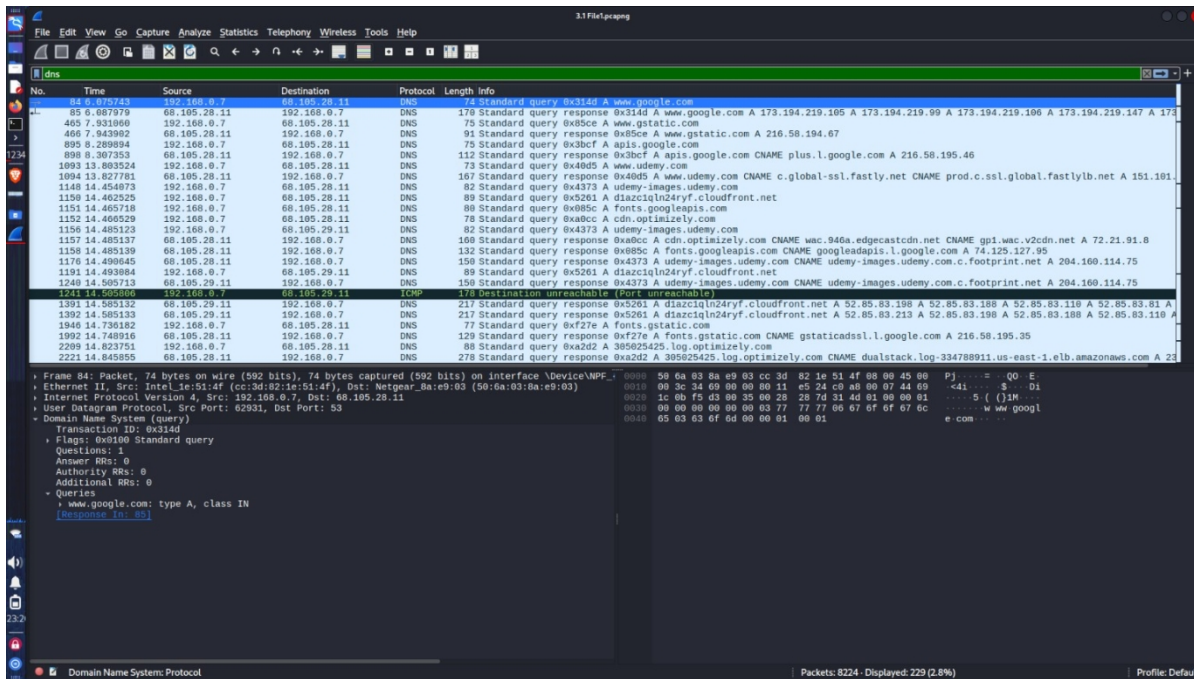


Figure 3. DNS query for www.google.com from the internal host.

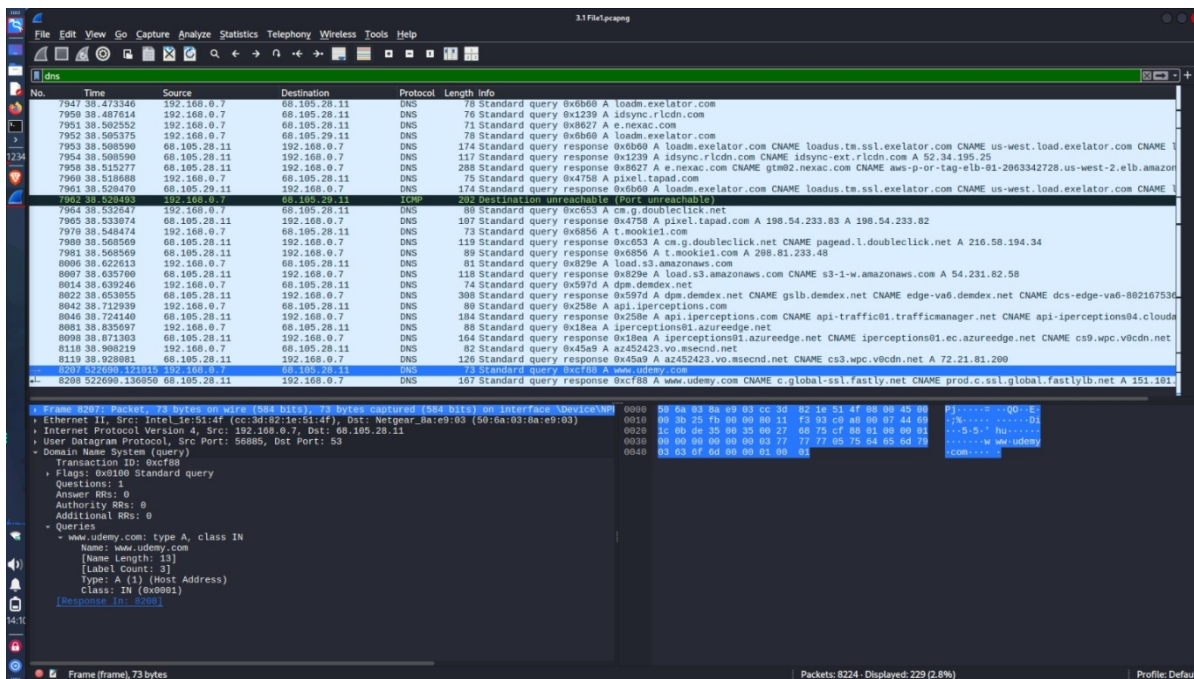


Figure 4. DNS query for www.udemy.com and the corresponding DNS response traffic.

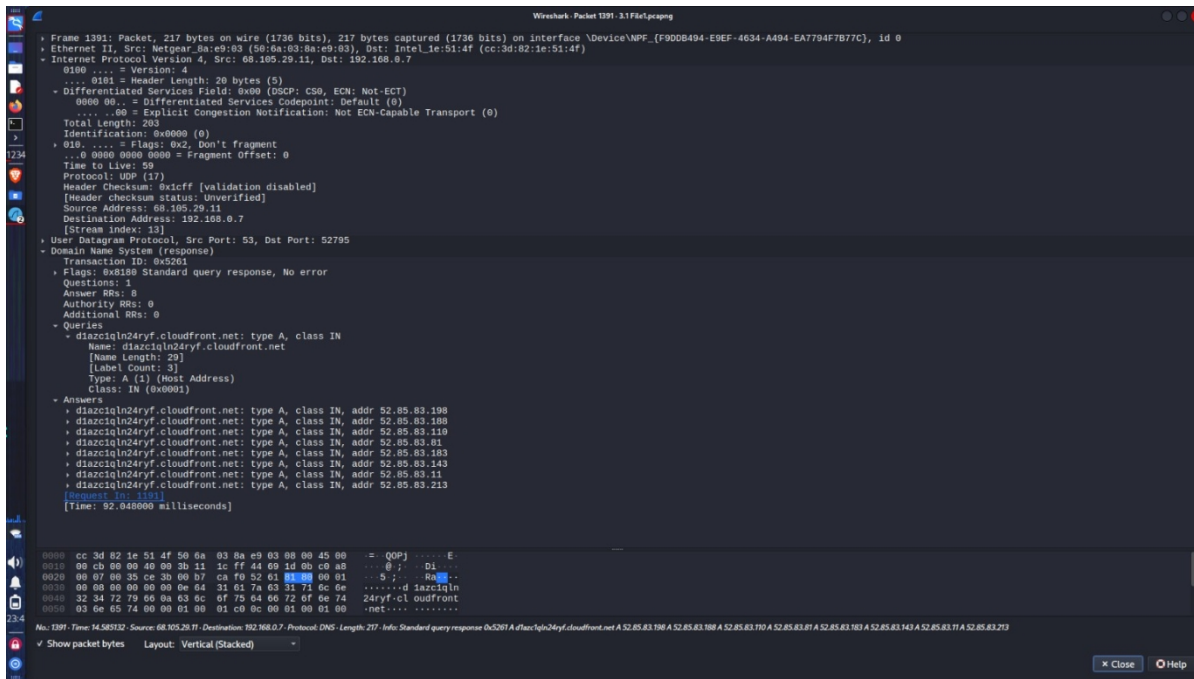


Figure 5. Multiple external destination addresses visible while filtering traffic from 192.168.0.7.

6. TCP Findings

The capture provides examples of normal TCP session establishment. A client begins with SYN, the server replies with SYN-ACK, and the client completes the handshake with ACK. The screenshots also show TCP connections involving port 443 and TLS Client Hello packets, demonstrating encrypted web-session establishment.

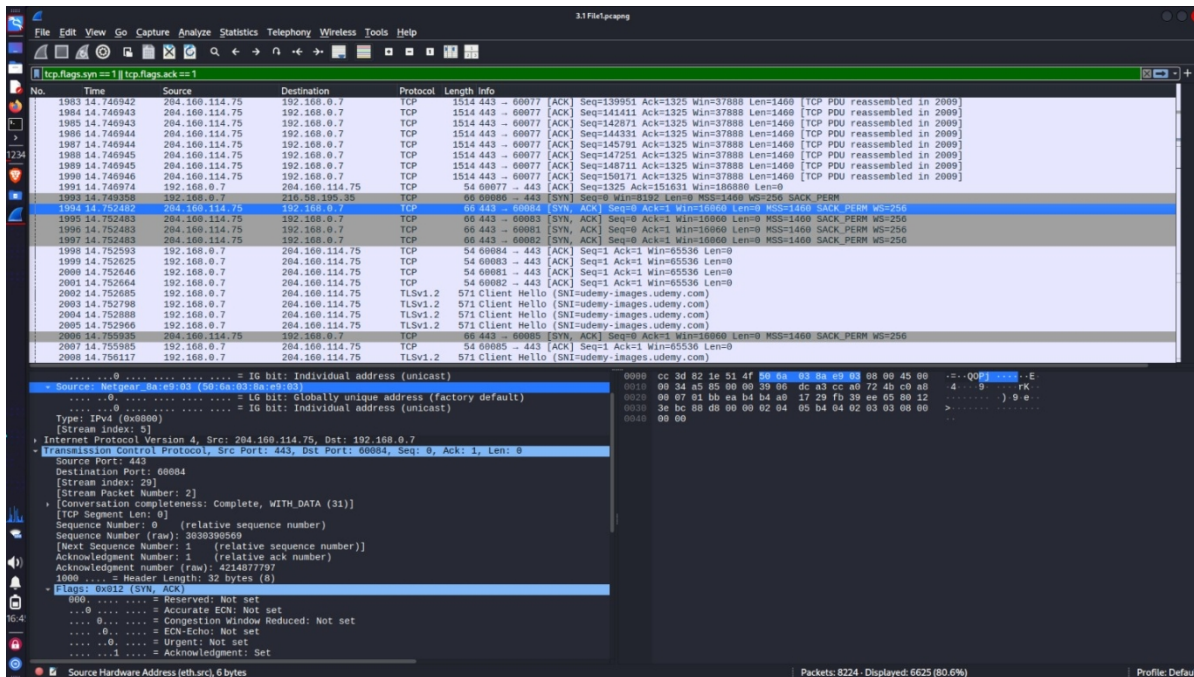


Figure 6. TCP traffic showing SYN/SYN-ACK establishment for a connection to port 443 and subsequent TLS Client Hello packets.

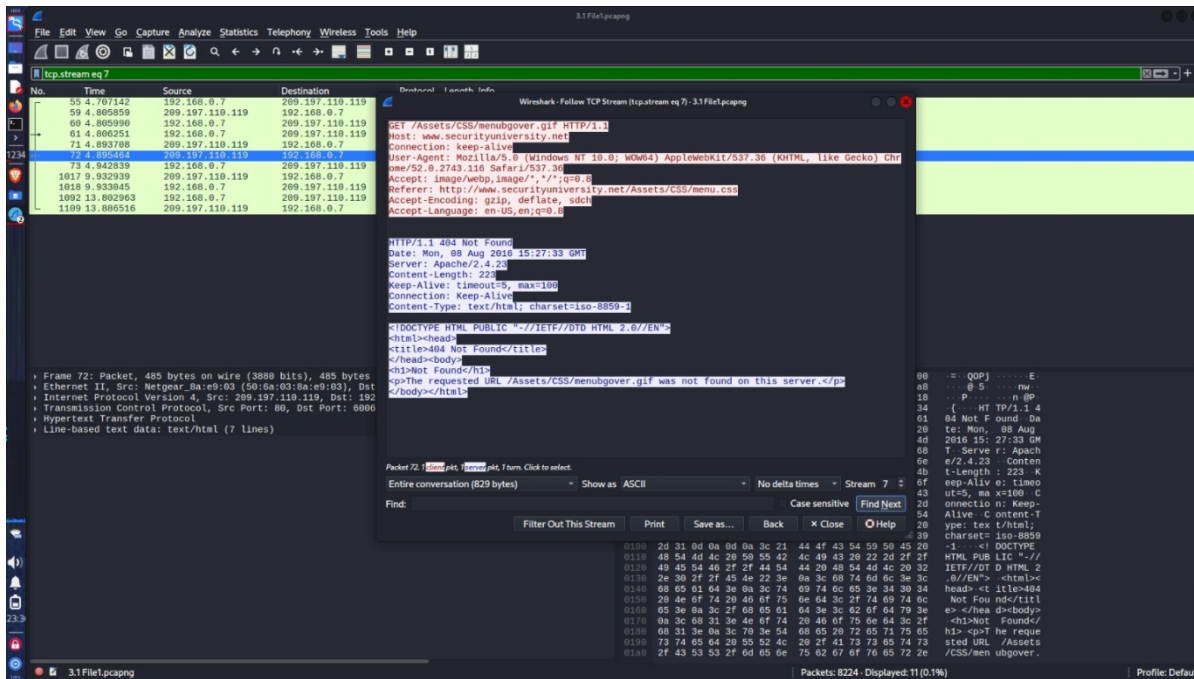


Figure 7. Follow TCP Stream view showing an HTTP request followed by an HTTP 302 response.

7. HTTP/HTTPS Findings

The HTTP evidence contains GET requests and server responses. One visible exchange requests /Assets/CSS/menubgover.gif and receives HTTP/1.1 404 Not Found. Another HTTP stream shows a request for /logging.php followed by HTTP/1.1 304 Not modified. These responses are useful for understanding application behaviour and troubleshooting.

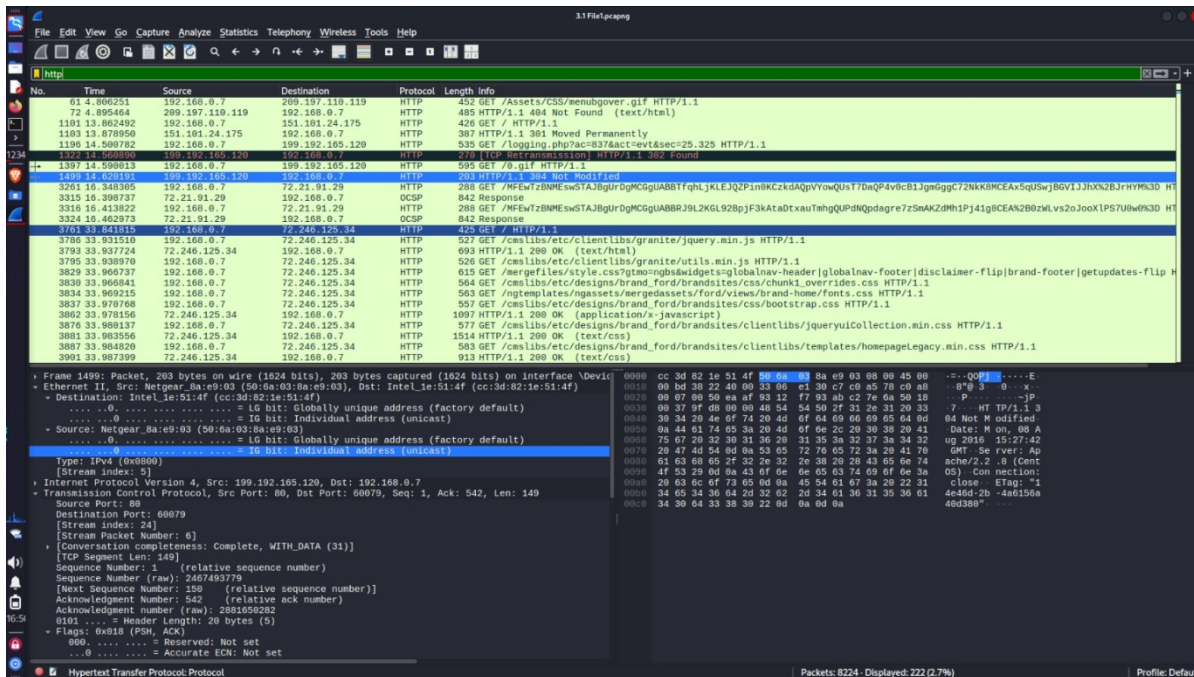


Figure 8. HTTP packet view showing GET requests and server responses, including 404 Not Found and 200 OK traffic.

Encrypted traffic is also visible. TLS/HTTPS protects application payloads, so a passive analyst cannot normally read the HTTP content directly. However, metadata such as source/destination addresses, ports, timing and TLS handshake information remains available for analysis.

8. Interesting Findings

Two useful investigation findings are visible in the evidence. First, the HTTP stream contains a 302 redirect and a 404 response, which are application-level events rather than proof of malicious activity. Second, the capture contains a TCP retransmission event. Retransmissions can result from packet loss, congestion or other network conditions and should therefore be investigated in context rather than automatically classified as an attack.

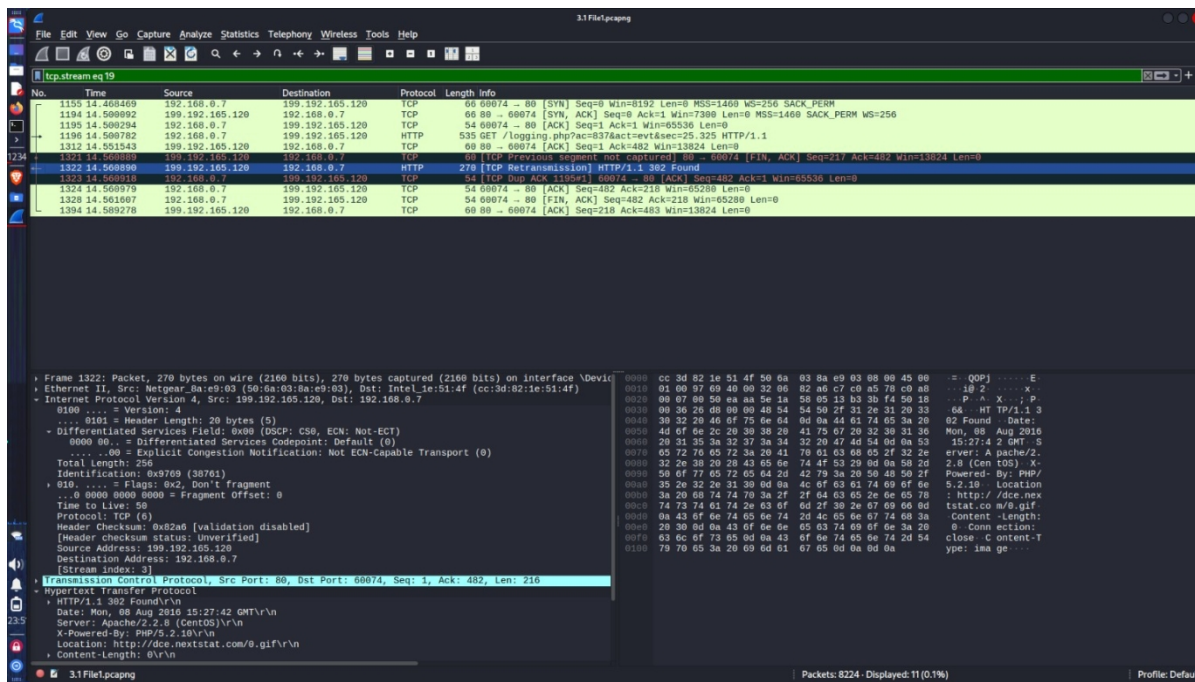


Figure 9. Wireshark view filtered for TCP retransmission activity.

The supplied capture also contains a DNS response for a CloudFront hostname returning multiple addresses. Multiple returned addresses can be normal for load balancing or content-delivery infrastructure and should not be treated as malicious without additional evidence.

9. Conclusion

The PCAP shows network activity generated by the internal host 192.168.0.7 across several common protocols. DNS resolves external hostnames, TCP establishes reliable sessions, HTTP exposes request/response information, and TLS/HTTPS or QUIC provides encrypted web communication. The evidence also demonstrates why packet analysis should be evidence-driven: redirects, 404 responses, retransmissions and multiple destination addresses are noteworthy events, but none alone proves malicious activity.

TASK 2 — IP ADDRESSES AND NETWORK HEADERS

2.1 What is an IP Address?

An Internet Protocol (IP) address is a numerical network address assigned to an interface or device so that IP packets can be delivered to the correct destination. At the network layer, the address identifies where traffic originates and where it is intended to go. IP addressing allows devices on interconnected networks to communicate even when they are located on different local networks.

2.2 Difference Between IPv4 and IPv6

IPv4 uses 32-bit addresses, normally written as four decimal octets such as 192.168.0.7. Its address space is approximately 4.3 billion addresses, although practical availability is lower because some ranges are reserved for special purposes. IPv6 uses 128-bit addresses written in hexadecimal groups separated by colons, for example 2001:db8::1. IPv6 was introduced primarily to provide a vastly larger address space and also defines a different packet-header structure and addressing model.

2.3 Public vs. Private IP Addresses

Public IP addresses are globally routable addresses used to identify networks or interfaces across the public Internet. Private IPv4 addresses are reserved for internal networks and are not directly routable across the public Internet. The main private IPv4 ranges are 10.0.0.0/8, 172.16.0.0/12 and 192.168.0.0/16. The observed host 192.168.0.7 therefore falls within the private IPv4 address space. Network Address Translation (NAT) is commonly used to allow private hosts to communicate with public Internet destinations.

2.4 IP Addresses in Networking and Cybersecurity

IP addresses support routing, network segmentation, access-control policies, firewall rules, monitoring and troubleshooting. In cybersecurity investigations, they help analysts associate traffic with sources and destinations, identify communication between internal and external systems, and reconstruct parts of an incident timeline. However, an IP address is not automatically the identity of a person because NAT, proxies, shared infrastructure and cloud services can place multiple users or systems behind the same address.

2.5 What is Network Header ?

Network header information is control and metadata carried at the beginning of a protocol data unit. Headers tell receiving systems how to interpret, process and deliver the packet or segment. In a packet capture, headers provide valuable evidence even when the application payload is encrypted.

2.6 Common Information Found in Network Headers

- Source and destination IP addresses — identify network endpoints.
- Protocol or Next Header — identifies the transport or upper-layer protocol such as TCP, UDP or ICMP.
- TTL in IPv4 or Hop Limit in IPv6 — limits packet lifetime across routing hops.
- TCP/UDP source and destination ports — identify application endpoints.
- TCP sequence and acknowledgement numbers — support reliable, ordered TCP communication.
- TCP flags such as SYN, ACK, FIN and RST — indicate connection state and control events.
- Packet length and related size information — helps assess traffic volume and unusual patterns.
- IPv4 identification, fragmentation flags and fragment offset — support analysis of fragmented packets.
- IPv6 traffic class, flow label and next-header information — provide IPv6-specific processing information.

2.7 How Header Information Supports Network Analysis and Security Investigations

Header information allows an analyst to reconstruct who communicated with whom, when communication occurred, which transport protocol was used and which service ports were involved. For example, SYN, SYN-ACK and ACK packets demonstrate normal TCP session establishment, while repeated connection attempts or unusual destination ports may warrant further investigation.

Headers remain valuable when payloads cannot be inspected. With encrypted TLS traffic, an analyst can still determine source and destination IP addresses, ports, packet sizes, timing and TCP behaviour. Combining these metadata with DNS queries, TLS information, authentication logs and endpoint telemetry can help distinguish expected activity from suspicious activity.

In the supplied PCAP, these principles can be seen directly: 192.168.0.7 is the internal host, 68.105.28.11 appears as the DNS server, and the capture includes TCP, UDP/QUIC, HTTP and TLS-related traffic. This demonstrates how header-level evidence supports both routine network analysis and security investigation.

Task 2 Sources

IETF RFC 791 — Internet Protocol, Version 4: <https://www.rfc-editor.org/rfc/rfc791>

IETF RFC 8200 — Internet Protocol, Version 6: <https://www.rfc-editor.org/rfc/rfc8200>

IANA — IPv4 Special-Purpose Address Space: <https://www.iana.org/assignments/iana-ipv4-special-registry/iana-ipv4-special-registry.xhtml>

IANA — IPv6 Special-Purpose Address Space: <https://www.iana.org/assignments/iana-ipv6-special-registry/iana-ipv6-special-registry.xhtml>

IETF RFC 1918 — Address Allocation for Private Internets: <https://www.rfc-editor.org/rfc/rfc1918>

IETF RFC 9293 — Transmission Control Protocol: <https://www.rfc-editor.org/rfc/rfc9293>

TASK 3 — FIVE MAJOR CYBERSECURITY INCIDENTS

The five cases below were selected to prioritize recent, high-impact incidents from 2020–2024 while ensuring that the required Nigerian #EndSARS/Anonymous case is included. They cover hacktivism, software supply-chain compromise, ransomware against critical infrastructure, social-engineering-led cyber extortion, and a healthcare ransomware incident with nationwide consequences. The discussion uses high-level attack mechanics and distinguishes confirmed attribution from public claims.

1. Anonymous / #OpNigeria / #EndSARS Cyberattacks

Year: 2020

Organization(s) affected: Nigerian government websites and online government services, including the Nigeria Police Force and Lagos State Government websites

Country/Region: Nigeria

What Happened

During the October 2020 #EndSARS protests, the decentralised hacktivist collective Anonymous publicly declared support for protesters and claimed a series of cyberattacks against Nigerian government targets. Reporting by Nigerian outlets documented disruption or unavailability affecting government websites, including the Nigeria Police Force website and the Lagos State Government website. Reports also described claims of data exposure and compromised social-media accounts. The episode is best understood as a hacktivist cyber campaign associated with the wider #EndSARS protest movement rather than as a conventional criminal ransomware operation.

How the Attack Was Carried Out

The campaign involved website disruption and compromise of online government assets, with Anonymous-associated accounts publicly claiming attacks. Some reporting described websites being taken offline and government information being exposed. The exact technical method was not consistently established in the public reporting reviewed, so this report does not label every incident as a confirmed DDoS or claim a single attack method for all affected sites.

Impact/Damage

The Nigeria Police Force website remained unavailable for days after the reported attack, forcing the police to rely on social-media channels for public communication. The Guardian also reported disruption affecting the Lagos State Government website and claims involving other government institutions. The wider significance was reputational and operational: government online services became targets during a period of intense civil unrest, showing how hacktivism can amplify political pressure.

How the Attackers Were Detected, Identified

Anonymous-affiliated accounts publicly claimed responsibility for several attacks, and Nigerian media reported those claims alongside observable website outages, however, the sources reviewed do not establish that specific individuals were conclusively identified, arrested or convicted for the attacks. The Central Bank of Nigeria, for example, publicly denied that its website had been hacked despite reports based on Anonymous' claims. Therefore, the strongest evidence is a combination of public claims and contemporaneous observations, not a confirmed identification of individual attackers.

Key Cybersecurity Lessons

- Treat hacktivist claims as intelligence that must be independently verified rather than as automatic proof of attribution.
- Protect public-facing government websites with layered availability and DDoS-resilience controls.
- Use multi-factor authentication for official social-media and administrative accounts.
- Maintain alternative communication channels so public services can continue during website outages.
- Preserve logs, DNS records, web-server evidence and other forensic data immediately after an incident.
- Separate verified technical facts from politically motivated claims when reporting a cyber incident.

Sources

The ICIR — #ENDSARS: One week after attack on Nigeria's police website: <https://www.icirnigeria.org/endsars-one-week-after-attack-on-nigerias-police-website-portal-still-inactive/>

The ICIR — Anonymous allegedly hacks Nigerian government website: <https://www.icirnigeria.org/endsars-international-hacking-group-anonymous-allegedly-hacks-nigerian-government-website/>

The Guardian Nigeria — Cyberattack hits Lagos State Government website: <https://guardian.ng/news/cyberattack-hits-lagos-state-government-website/>

The Guardian Nigeria — Anonymous claims hacking of State House websites: <https://guardian.ng/news/anonymous-claims-hacking-of-state-house-websites-in-support-of-endsars/>

The ICIR — CBN denies reports its website was hacked: <https://www.icirnigeria.org/cbn-denies-reports-its-website-was-hacked/>

2. SolarWinds Orion / SUNBURST Supply-Chain Compromise

Year: 2020

Organization(s) affected: SolarWinds and selected government agencies and private-sector organizations worldwide

Country/Region: Global; major U.S. government impact

What Happened

Attackers compromised the software development and build process used for SolarWinds Orion and inserted the SUNBURST backdoor into legitimate, digitally signed software updates released during 2020. Customers that installed affected versions could unknowingly introduce the malicious component into trusted environments. The operation became one of the most important examples of a modern software supply-chain compromise.

How the Attack Was Carried Out

The attackers first gained access to SolarWinds' environment and manipulated the Orion build process. Malicious code was inserted into legitimate Orion software builds and distributed through normal update channels. Once activated in a victim environment, SUNBURST communicated with attacker-controlled infrastructure and could support further post-compromise activity. The technique was particularly effective because the initial software package was trusted and digitally signed.

Impact/Damage

SolarWinds initially reported that up to 18,000 customers had downloaded potentially affected Orion versions, but later estimated that fewer than 100 customers were actually hacked through SUNBURST. The campaign nevertheless affected major government and private-sector organizations and demonstrated how compromise of one software supplier can create systemic risk across many customers.

How the Attackers Were Detected, Identified

FireEye/Mandiant publicly disclosed the supply-chain compromise in December 2020 after its investigation into a separate intrusion revealed the SUNBURST backdoor. Technical analysis of the malware, command-and-

control infrastructure and related activity enabled researchers and government agencies to connect the campaign to a sophisticated nation-state actor. In April 2021, the U.S. government formally attributed the SolarWinds supply-chain compromise to Russia's Foreign Intelligence Service (SVR).

Key Cybersecurity Lessons

- Treat software suppliers and update mechanisms as part of the organization's security boundary.
- Secure and monitor software build and release pipelines.
- Use code-signing protections and verify build integrity rather than trusting software solely because it is digitally signed.
- Monitor trusted applications for anomalous network behaviour.
- Maintain detailed logs and threat-hunting capability for long-lived, low-noise intrusions.
- Use network segmentation and least privilege to limit post-compromise movement.

Sources

CISA — Active Exploitation of SolarWinds Software: <https://www.cisa.gov/news-events/alerts/2020/12/13/active-exploitation-solarwinds-software>

CISA — Russian SVR Targets U.S. and Allied Networks: <https://www.cisa.gov/news-events/alerts/2021/04/15/nsa-cisa-fbi-joint-advisory-russian-svr-targeting-us-and-allied>

Mandiant/Google Cloud — SUNBURST Additional Technical Details: <https://cloud.google.com/blog/topics/threat-intelligence/sunburst-additional-technical-details/>

SolarWinds — Investigative Update of the Cyberattack: <https://www.solarwinds.com/blog/an-investigative-update-of-the-cyberattack>

SolarWinds SEC filing — Investigative Update: <https://www.sec.gov/Archives/edgar/data/1739942/000173994221000080/swi-20210331.htm>

3. Colonial Pipeline Ransomware Attack

Year: 2021

Organization(s) affected: Colonial Pipeline Company

Country/Region: United States

What Happened

On 7 May 2021, Colonial Pipeline suffered a ransomware attack attributed by the FBI to the DarkSide ransomware group. Colonial proactively shut down its pipeline system after the compromise. The shutdown lasted several days and affected fuel distribution across parts of the eastern United States.

How the Attack Was Carried Out

The incident was a ransomware intrusion into Colonial's information-technology environment. DarkSide used compromised access to encrypt or otherwise disrupt systems and demand payment. Colonial shut down pipeline operations as a precaution while it investigated and contained the incident. Public reporting and government statements indicate that the attack primarily affected IT systems used by the company; the pipeline's operational technology was not itself reported as directly compromised.

Impact/Damage

Colonial restarted its entire pipeline system on 13 May 2021 after a five-day shutdown. The disruption contributed to fuel shortages and widespread public concern along the U.S. East Coast. Colonial paid a ransom of approximately 75 bitcoin. The U.S. Department of Justice later seized about 63.7 bitcoin, then valued at approximately \$2.3 million, from a cryptocurrency wallet associated with the ransom proceeds.

How the Attackers Were Detected, Identified

The FBI confirmed on 10 May 2021 that DarkSide was responsible for the compromise. Investigators then followed the ransom through the Bitcoin public ledger. According to the DOJ, approximately 63.7 bitcoin representing ransom proceeds were transferred to a specific address for which the FBI obtained the private key, enabling seizure of the funds. The case therefore demonstrates attribution to a ransomware group plus a concrete financial-tracing and seizure outcome, even though public sources did not announce the arrest of the individual attacker who personally entered Colonial's network.

Key Cybersecurity Lessons

- Critical infrastructure operators need ransomware-resistant segmentation between IT and operational environments.
- Maintain tested offline or otherwise resilient backups and recovery plans.
- Establish rapid incident-reporting relationships with law enforcement before an incident occurs.
- Monitor privileged and remote access for unusual activity.
- Cryptocurrency payments can leave traceable financial evidence on public blockchains.
- Business continuity planning is essential because IT disruption can produce physical and economic consequences.

Sources

FBI — Statement on Compromise of Colonial Pipeline Networks: <https://www.fbi.gov/news/press-releases/fbi-statement-on-compromise-of-colonial-pipeline-networks>

U.S. DOJ — Seizure of \$2.3 Million in Cryptocurrency Paid to DarkSide: <https://www.justice.gov/archives/opa/pr/departments-justice-seizes-23-million-cryptocurrency-paid-ransomware-extortionists-darkside>

U.S. Department of Energy — Colonial Pipeline Cyber Incident: <https://www.energy.gov/ceser/colonial-pipeline-cyber-incident>

FBI — Remarks on Colonial Pipeline ransomware response: <https://www.fbi.gov/news/press-releases/fbi-deputy-director-paul-abbates-remarks-at-press-conference-regarding-the-ransomware-attack-on-colonial-pipeline>

4. MGM Resorts Cyberattack

Year: 2023

Organization(s) affected: MGM Resorts International

Country/Region: United States

What Happened

In September 2023, MGM Resorts identified a cybersecurity incident involving unauthorized access to parts of its U.S. systems. The company shut down systems to contain the intrusion, but the defensive action caused major disruption across hotel and casino operations. Personal information belonging to some customers was also obtained.

How the Attack Was Carried Out

Public threat-intelligence reporting links the attack pattern to Scattered Spider/UNC3944, a financially motivated cluster known for telephone-based social engineering, SMS phishing, credential theft and SIM-swapping. The attackers were able to exploit weaknesses in identity and help-desk processes to gain access to corporate accounts. After access was obtained, the attackers disrupted systems and conducted data theft/extortion activity. This report keeps the mechanics at a high level and does not reproduce operational attack steps.

Impact/Damage

MGM estimated that the September disruption would reduce adjusted property EBITDAR by approximately \$100 million, mainly affecting its Las Vegas operations, and reported less than \$10 million in one-time incident-response expenses in the third quarter. Customer information obtained included names, contact details, gender, dates of birth and driver's-license numbers; for a limited number of customers, Social Security and passport numbers were also affected. MGM said it did not believe bank-account numbers or payment-card information were obtained.

How the Attackers Were Detected, Identified, and Caught

MGM detected the unauthorized activity and began an investigation with external cybersecurity experts and law enforcement. Mandiant's public reporting identifies UNC3944 as overlapping with the activity publicly known as Scattered Spider, and later FBI reporting described Scattered Spider as a highly capable social-engineering group. In 2026, the U.S. Department of Justice announced the arrest and extradition of an alleged Scattered Spider member, Peter Stokes. However, that later prosecution should not be presented as proof that Stokes personally carried out the MGM intrusion. The defensible conclusion is that the campaign has been associated with the Scattered Spider/UNC3944 ecosystem, while individual responsibility for the MGM attack was not publicly established in the sources reviewed.

Key Cybersecurity Lessons

- Help-desk identity verification is a critical security control, not merely an administrative function.
- Use phishing-resistant MFA where practical and protect account-recovery processes.
- Monitor unusual authentication, SIM changes and privileged-account activity.
- Prepare manual operating procedures for hotels, casinos and other high-availability businesses.
- Segment sensitive systems so one compromised identity cannot reach every critical service.
- Incident-response plans should anticipate both operational disruption and data-extortion pressure.

Sources

Mandiant/Google Cloud — UNC3944 SMS phishing, SIM swapping and extortion: <https://cloud.google.com/blog/topics/threat-intelligence/unc3944-sms-phishing-sim-swapping-ransomware>

MGM Resorts SEC filing — September 2023 cybersecurity issue:

<https://www.sec.gov/Archives/edgar/data/789570/000119312523251667/d461062d8k.htm>

MGM Resorts — Cybersecurity issue update: <https://investors.mgmresorts.com/2023-10-05-MGM-RESORTS-UPDATE-ON-RECENT-CYBERSECURITY-ISSUE>

FBI — Scattered Spider threat intelligence discussion: <https://www.fbi.gov/video-repository/ahead-of-the-threat-s1-ep-5-rachel-lavender/view>

U.S. DOJ — Alleged Scattered Spider member arrested and extradited, 2026: <https://www.justice.gov/opa/pr/alleged-member-criminal-cyber-hacking-group-scattered-spider-arrested-finland-and-extradited>

5. Change Healthcare Ransomware Attack

Year: 2024

Organization(s) affected: Change Healthcare / UnitedHealth Group and connected healthcare providers

Country/Region: United States

What Happened

On 21 February 2024, Change Healthcare, a major healthcare claims and payment intermediary owned by UnitedHealth Group, identified that cybercrime actors had gained access to its IT systems. The company isolated affected systems, which caused a major interruption to healthcare claims processing, payments and related services across the United States. ALPHV/BlackCat publicly claimed responsibility.

How the Attack Was Carried Out

The incident involved ransomware and data-extortion activity. Public reporting and congressional testimony linked the attack to ALPHV/BlackCat. The attack disrupted Change Healthcare's transaction-processing infrastructure, which is deeply embedded in the U.S. healthcare ecosystem. The exact initial-access details were not fully disclosed by UnitedHealth in its early SEC filings, so this report does not invent a specific entry method.

Impact/Damage

The consequences were unusually broad because Change Healthcare sits between healthcare providers and payers. UnitedHealth reported that it provided more than \$9 billion in interest-free loans to support providers through the end of 2024, incurred approximately \$2.2 billion in direct response costs during 2024 and experienced an estimated \$867 million in business-disruption impacts. UnitedHealth ultimately estimated that approximately 190 million individuals were affected by the data breach. The attack therefore affected both availability of essential healthcare transactions and confidentiality of personal/health information.

How the Attackers Were Detected, Identified

UnitedHealth detected the intrusion on 21 February and immediately isolated the affected systems. The attacker was publicly associated with ALPHV/BlackCat, which claimed responsibility. The U.S. government had already disrupted BlackCat infrastructure in December 2023 and developed a decryption tool for victims, but the sources reviewed do not establish that a specific individual was arrested for carrying out the Change Healthcare intrusion. Reports of a \$22 million ransom payment were based on cryptocurrency and criminal-forum evidence; UnitedHealth did not publicly confirm the payment, so it is treated here as reported rather than established fact.

Key Cybersecurity Lessons

- Critical healthcare intermediaries require resilience because a single supplier can affect thousands of downstream organizations.
- Protect remote access and privileged accounts with strong authentication and continuous monitoring.
- Maintain contingency channels for claims, payments and other essential healthcare transactions.
- Segment high-value systems and rehearse isolation procedures so containment does not become uncontrolled service failure.
- Treat data-extortion risk separately from encryption risk because attackers may steal data even when systems are restored.
- Coordinate quickly with government, healthcare partners, security researchers and law enforcement.

Sources

HHS — Change Healthcare Cybersecurity Incident FAQ: <https://www.hhs.gov/hipaa/for-professionals/special-topics/change-healthcare-cybersecurity-incident-frequently-asked-questions/index.html>

UnitedHealth Group — March 2024 Change Healthcare cyberattack update: <https://www.unitedhealthgroup.com/newsroom/2024/2024-03-07-uhg-update-change-healthcare-cyberattack.html>

UnitedHealth Group — April 2024 data-impact update: <https://www.unitedhealthgroup.com/newsroom/2024/2024-04-22-uhg-updates-on-change-healthcare-cyberattack.html>

UnitedHealth Group 2024 Form 10-K / SEC filing: <https://www.sec.gov/Archives/edgar/data/731766/000073176625000063/unh-20241231.htm>

Reuters — U.S. bounty and Change Healthcare/BlackCat reporting: <https://www.reuters.com/technology/cybersecurity/us-offers-10-million-bounty-info-blackcat-hackers-who-hit-unitedhealth-2024-03-27/>

U.S. DOJ — BlackCat disruption and decryption tool: <https://www.justice.gov/archives/opa/pr/justice-department-disrupts-prolific-alphvblackcat-ransomware-variant>

FINAL COMPARISON

Incident	Year	Main Attack Type	Main Impact	Attribution / Outcome
Anonymous / #EndSARS	2020	Hactivism; website disruption/compromise	Government websites disrupted; information reportedly exposed	Anonymous claims + observed outages; no conclusive individual identification established
SolarWinds / SUNBURST	2020	Supply-chain compromise / cyber espionage	Trusted software updates used to reach selected government and private-sector victims	U.S. government attributed campaign to Russian SVR; technical evidence supported attribution
Colonial Pipeline	2021	Ransomware	Five-day pipeline shutdown; fuel-supply disruption; ransom proceeds partly seized	FBI attributed to DarkSide; DOJ traced and seized about \$2.3m in bitcoin
MGM Resorts	2023	Social engineering / credential compromise / extortion	Major casino and hotel disruption; ~ \$100m EBITDAR impact; customer data exposed	Linked to Scattered Spider/UNC3944 ecosystem; later individual arrest did not prove responsibility for MGM
Change Healthcare	2024	Ransomware / data extortion	Nationwide healthcare transaction disruption; major financial and data impact	ALPHV/BlackCat claimed responsibility; no individual conviction established for this incident in reviewed sources

RESEARCH NOTE

Claims are presented conservatively, where public reporting or an attacker claim is not independently confirmed, the report explicitly identifies it as a claim rather than established fact. Taken together, the five incidents show that major cyberattacks begin through very different attack vectors. The #EndSARS campaign demonstrates how hactivists can use disruption and public claims to amplify political activity. SolarWinds shows the systemic danger of compromising a trusted software supplier rather than attacking every victim directly. Colonial Pipeline and Change Healthcare demonstrate how ransomware against a connected organization can create consequences far beyond the victim's own IT environment. MGM Resorts highlights a different weakness: sophisticated technical controls can be undermined when attackers successfully manipulate identity, authentication and help-desk processes.

The cases also demonstrate that detection and attribution are separate problems. A security team may detect unauthorized access quickly without knowing who is responsible, while intelligence agencies may later attribute an operation through technical indicators, infrastructure, malware analysis and intelligence. SolarWinds produced a formal nation-state attribution, while Colonial produced both group attribution and cryptocurrency seizure. In the Nigerian #EndSARS case, however, public claims by Anonymous and observable website disruption should not be confused with proof that particular individuals were identified or arrested.

Across the incidents, recurring defensive priorities include strong identity controls, phishing-resistant authentication, network segmentation, vulnerability and supply-chain management, continuous monitoring, reliable logging, resilient backups and tested incident-response plans. Organizations should also assume that third-party dependencies can become attack paths. Security monitoring must therefore cover trusted software, cloud services, help desks, remote access, DNS and network metadata rather than focusing only on traditional malware signatures, that said, cybersecurity resilience is not only about preventing compromise it is also about maintaining essential services during an incident, preserving evidence for investigation, communicating accurately with security engineers and cooperating with law enforcement.